Solution

A1. From thermodynamics we know that $v = \sqrt{3kT/m}$, where m is the particle's mass and k is the Boltzmann constant. The characteristic jump time between inter-nodes is inversely proportional to the transition probability times the particle's speed, and the diffusion coefficient is inversely proportional to the jump time; therefore

$$\boxed{v = \sqrt{\frac{3kT}{m}}, \qquad D(T) \propto v \exp(-E_g/kT) \propto \sqrt{T}\exp(-E_g/kT)}$$

A2. First, let us find the rms particle displacement along one of the axes (call it x). The motion along it happens as in the one considered in the preparatory lectures, so the mean square displacement is

$$\overline{x^2} = na^2,$$

where n is the total number of jumps. The number of jumps can be estimated as $n = t/\tau$. With $D = a^2/2\tau$ we obtain

$$\overline{x^2} = 2Dt.$$

Finally, because the motions along different axes can be considered independent, $\overline{y^2} = \overline{z^2} = 2Dt$. By the Pythagorean theorem,

$$\boxed{\Delta r_{sq} = \sqrt{\overline{x^2} + \overline{y^2} + \overline{z^2}} = \sqrt{6Dt}}$$

A3. Let us use the result of the previous task to find the diffusion coefficient. Because $\Delta r_{sq}^2 = 6Dt$, $D = \Delta r_{sq}^2/6t$. Combining this with A1, $\Delta r_{sq}^2/(6t\sqrt{T}) \propto \exp(-E_g/kT)$. The easiest way now is to find the slope of the dependence of $Y = \ln\left(\Delta r_{sq}^2 t^{-1} T^{-1/2}\right)$ on $X = 1/T$.

How the accuracy is guaranteed. First, it is good to have $\Delta r_{sq} \gg 1$. Second, $1/T$ changes by 0.9, and E_g is proportional to the slope. To guarantee $2\,\%$ accuracy the errors at the minimum and maximum temperatures must satisfy $\sigma_Y(0.1) + \sigma_Y(1.0) \le 0.02 \cdot 0.9 E_g$. Because $Y = \ln[\Delta r_{sq}^2 T^{-1/2}]$ (up to a constant), this means $2\varepsilon_r(0.1) + 2\varepsilon_r(1.0) \le 0.018 E_g$, i.e. it suffices that $\varepsilon_r = \sigma_r/\Delta r_{sq} \le 0.0045 E_g$ at every point. The data points:

t	10	10	10	10	10	10	10	10	10	10
T	1.000	1.111	1.250	1.429	1.667	2.000	2.500	3.333	5.000	10.00
N	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000
Δr_{sq}	13.750	17.009	20.882	25.999	32.154	40.362	50.733	65.932	86.227	122.632
σ_r	0.053	0.066	0.081	0.102	0.127	0.156	0.198	0.257	0.336	0.477
X	1.000	0.900	0.800	0.700	0.6	0.5	0.4	0.3	0.2	0.1
Y	2.939	3.312	3.664	4.035	4.383	4.747	5.092	5.473	5.807	6.164

One can see that the required accuracy is indeed achieved. Plotting $Y(X)$ and finding its slope with `InteractiveBuffer.nb`:



	1	2
1	1.	2.939
2	0.9	3.312
3	0.8	3.664
4	0.7	4.035
5	0.6	4.383
6	0.5	4.747
7	0.4	5.092
8	0.3	5.473
9	0.2	5.807
10	0.1	6.164

The fit of $Y(X)$ in `InteractiveBuffer.nb`.

$$-E_g = -3.58 \pm 0.02 \quad \Rightarrow \quad E_g = 3.58 \pm 0.02$$

> Editor's note: the question paper numbers this task A2 a second time; the answer sheet and this solution call it A3.

B1. From the question paper we can conclude that the grain boundaries perpendicular to the diffusion flow do not affect the effective diffusion coefficient. Of the three families of boundaries (total relative volume δ), the two parallel to the diffusion direction carry $2\delta/3$, so

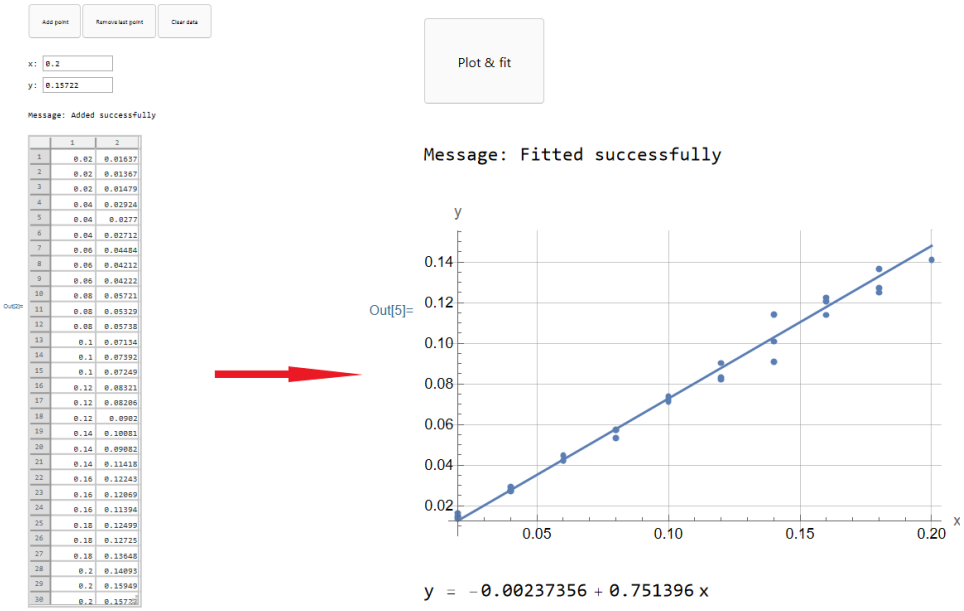
$$D_{eff} \approx \frac{2\delta D_{gb}}{3} \quad \Rightarrow \quad \alpha = \frac{2}{3}$$

B2. The simulation results are better for larger D_{gb}/D and smaller δ. It is also better to use the maximum simulation time, since type A diffusion appears on large time scales. So let us take $(t, D_{gb}/D, \delta) = (5, 5, 0.005)$ as the starting point. The plots in `B.nb` should ideally have the slope $k = -1/(4D_{eff}t)$, so in the program's units

$$D_{eff} = \frac{1}{4|k|t} = 1 + \alpha\delta(D_{gb}/D - 1).$$

With the coordinates $X = \delta(D_{gb}/D - 1)$ and $Y = \dfrac{1}{4|k|t} - 1$ this dependence is linear. The best way to make X change considerably is to increase δ. Lowering the *Width* slider in `B.nb` also helps against edge effects. The data points ($t = 5$, $D_{gb}/D = 5$):

δ	X	$\lvert k\rvert$ **(three runs)**			Y		
0.005	0.02	0.049194	0.049325	0.049271	0.01637	0.01367	0.01479
0.010	0.04	0.048579	0.048652	0.048679	0.02924	0.02770	0.02712
0.015	0.06	0.047854	0.047978	0.047974	0.04484	0.04212	0.04222
0.020	0.08	0.047294	0.047470	0.047286	0.05721	0.05329	0.05738
0.025	0.10	0.046670	0.046558	0.046616	0.07134	0.07392	0.07259
0.030	0.12	0.046159	0.046208	0.045863	0.08321	0.08206	0.09020
0.035	0.14	0.045421	0.045837	0.044876	0.10081	0.09082	0.11418
0.040	0.16	0.044546	0.044615	0.044885	0.12243	0.12069	0.11394
0.045	0.18	0.044445	0.044355	0.043995	0.12499	0.12725	0.13648
0.050	0.20	0.043823	0.043122	0.043207	0.14093	0.15949	0.15722



The fit of $Y(X)$ in `InteractiveBuffer.nb`.

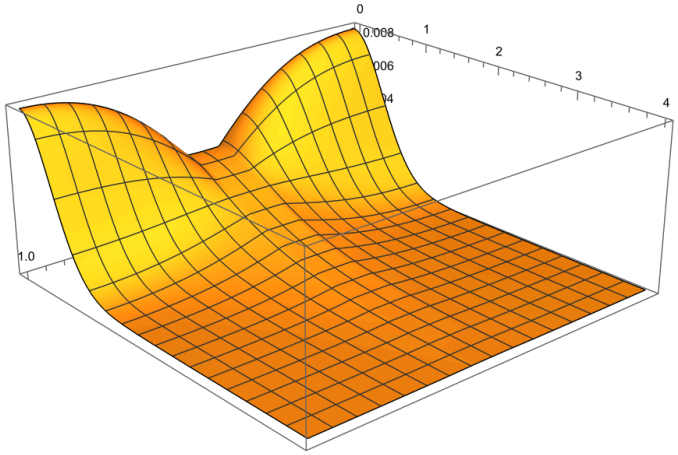
$$\alpha = 0.75 \pm 0.09$$

This coincides with the theoretical estimate of B1 within the margin of error.

C1. The only a priori constraint here is the dimensional one. Every factor on the right-hand side except a is dimensionless, so

$$\psi = -\frac{6}{5}$$

C2. As stated in the question paper, the scale of type B diffusion must be much smaller than a (i.e. $\sqrt{2[D_{gb}/D]t} \ll 1$) and at the same time much larger than δa (i.e. $\sqrt{2t} \gg \delta$). If we take, for example, the input parameters $(t, D_{gb}/D, \delta) = (0.100, 25, 0.10)$, we can obtain type B diffusion while varying any of the input parameters in a wide range. Whether the simulated diffusion was of type B can always be checked by plotting the concentration profile and comparing it with the one in the question paper. For example, the concentration map for $(0.100, 25, 0.10)$:



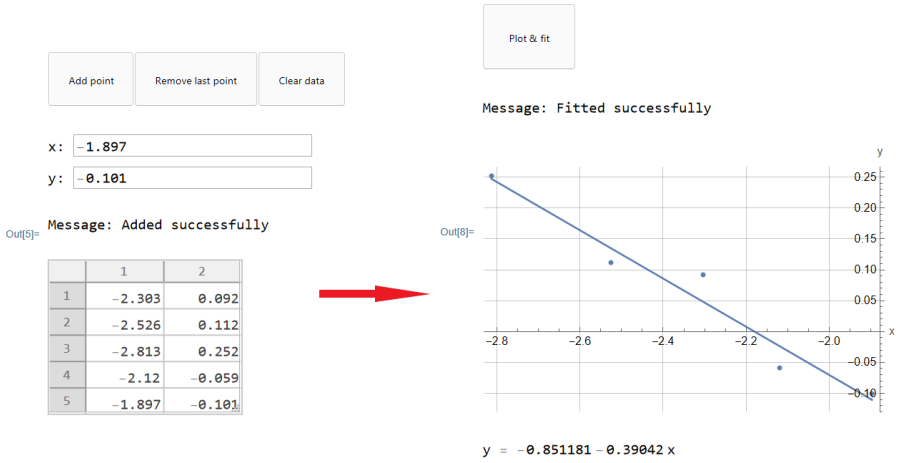
Concentration map for $(t, D_{gb}/D, \delta) = (0.100, 25, 0.10)$.

To find the slope $k = d\ln\bar{c}/dz^{6/5}$ we clip the linear regions of the plots generated in `C.nb`. ξ is obtained by simulating diffusion with different δ and plotting $\ln|k|(\ln\delta)$ in `InteractiveBuffer.nb`; η and ζ are found in the same way, varying D_{gb}/D and t. The data points (with $Z \equiv \delta^{-0.39}[D_{gb}/D]^{-0.68}t^{-0.44}$, used in C3):

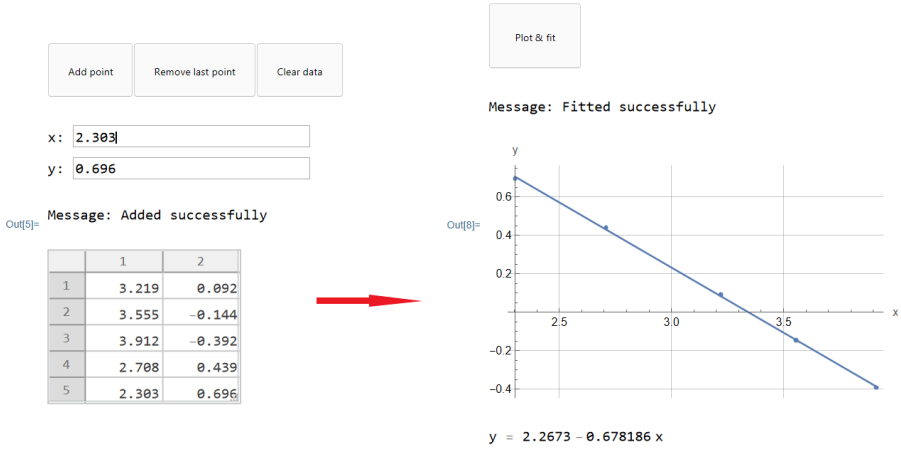
δ	0.100	0.080	0.060	0.120	0.150
k	−1.08567	-1.11796	-1.2865	-0.942494	-0.903676
$\ln\delta$	-2.303	-2.526	-2.813	-2.120	-1.897
$\ln\|k\|$	0.092	0.112	0.252	-0.059	-0.101
Z	0.758	0.826	0.925	0.706	0.647

D_{gb}/D	25	35	50	15	10
k	−1.08567	-0.865926	-0.675888	-1.55123	-2.00586
$\ln[D_{gb}/D]$	3.219	3.555	3.912	2.708	2.303
$\ln\|k\|$	0.092	-0.144	-0.392	0.439	0.696
Z	0.758	0.603	0.473	1.072	1.413

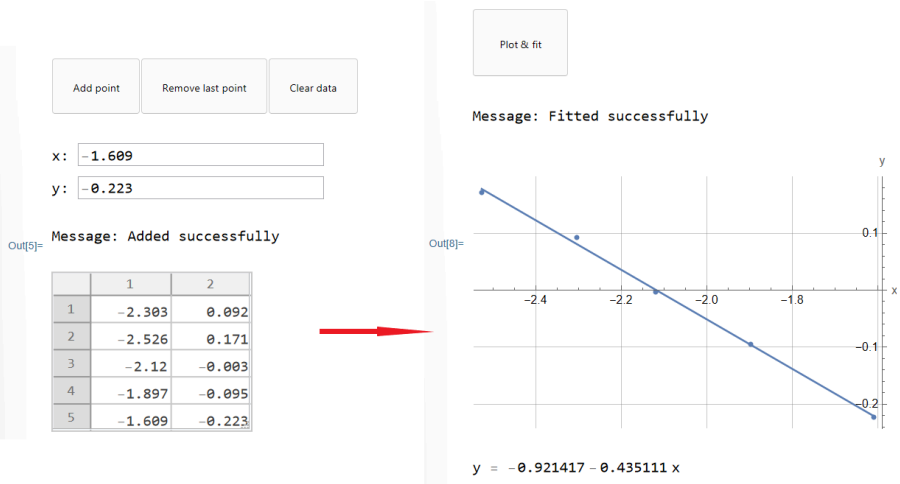
t	0.100	0.080	0.120	0.150	0.200
k	−1.08567	-1.18623	-0.99746	-0.908966	-0.799996
$\ln t$	-2.303	-2.526	-2.120	-1.897	-1.609
$\ln\|k\|$	0.092	0.171	-0.003	-0.095	-0.223
Z	0.758	0.836	0.699	0.634	0.558



$\ln|k|$ against $\ln\delta$.



$\ln|k|$ against $\ln[D_{gb}/D]$.

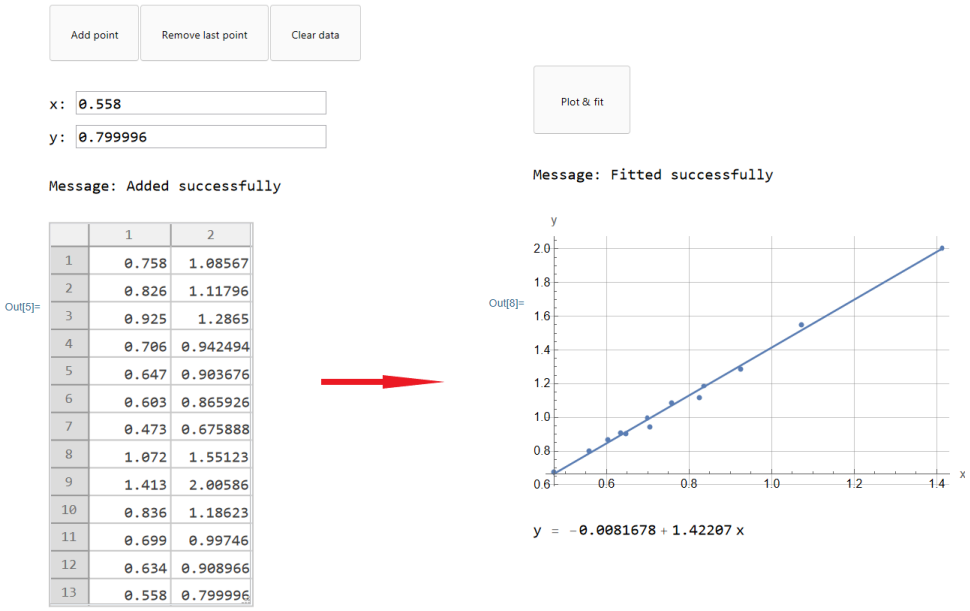


$\ln|k|$ against $\ln t$.

The coefficients we are looking for are the slopes of these lines:

$$\xi = -0.39, \qquad \eta = -0.68, \qquad \zeta = -0.44$$

C3. Knowing ψ, ξ, η and ζ, we find C as the slope of the dependence of $Y = |k|$ on $X = Z = \delta^{-0.39}[D_{gb}/D]^{-0.68}t^{-0.44}$. Calculating X for the points of the previous task, we plot it with `InteractiveBuffer.nb`:



	1	2
1	0.758	1.08567
2	0.826	1.11796
3	0.925	1.2865
4	0.706	0.942494
5	0.647	0.903676
6	0.603	0.865926
7	0.473	0.675888
8	1.072	1.55123
9	1.413	2.00586
10	0.836	1.18623
11	0.699	0.99746
12	0.634	0.908966
13	0.558	0.799996

The fit of $|k|$ against Z.

$$\boxed{C = 1.42}$$